Interaction of Beta Particles and X-Rays in Matter

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This experiment was performed in collaboration with Konrad Malirz.

Abstract

The aim of this experiment was to investigate the response of the X100-7 silicon photodiode to beta particles, X-rays and gamma rays. The photodiode was exposed to gamma radiation from Americium-241 (Am-241), Caesium-137 (Cs-137) and Sodium-22 (Na-22). Major features of the decay spectrum for each radioactive source were identified from the graphs generated by the photodiode, including photopeaks, the Compton Edge and the Compton background. The known energies for the compton edge of Sodium-22 and the photopeak of Am-241 were used to perform an energy calibration. The calibration was found to be linear, each channel corresponds to 2.65 ± 0.08 KeV with a low energy offset of -4.12 ± 0.02 KeV.

1. Introduction

Understanding the interactions of X-rays and gamma rays with matter is useful because the same interactions occur when photons interact with tissue in the human body. The properties of the interaction can be used to produce an image of the human body, as the rays pass through soft tissue relatively unimpeded. [1] The rays then pass onto a photographic plate or digital recorder, producing a radiograph. A 3-D image can be produced using computerised tomography (CT), by rotating the source of radiation and a detector around a patient's body [2]. Higher energy X-rays and gamma rays are also used in radiation treatment for cancer. The photons from these rays damage the DNA of cells, especially those that are in the process of mitosis, whereby a single cell splits into two new cells. Since cancerous cells divide quickly they are the most affected by the radiation. Early stage cancer can be cured by radiation as the tumor will shrink when the cancerous cells stop reproducing and die out over time. Radiation therapy is also used to prevent cancer recurring or spreading to other parts of the body [3].

2. Theory

2.1 Beta Decay

Beta decay occurs when an atom has an excess of neutrons or protons, producing an electron or positron respectively via the weak interaction. In β^- decay a neutron turns into a proton through the emission of a virtual W^- boson. W^- emission turns a down quark into an up quark, meaning a neutron, consisting of an up quark and two down quarks, is converted into a proton, consisting of two up quarks and one down quark. The W^- boson then decays into an electron and an antineutrino. In β^+ decay a proton is converted into a neutron, a positron and an electron neutrino. Beta particles have a mass around a thousandth of the size of a proton and so can reach reach relativistic speeds. Their light mass means they lose energy quickly through interactions with matter. Beta particles also undergo elastic scattering off nuclei, which can significantly change the direction of motion, meaning they follow a zig-zag path through absorbing material. This means the path that beta particles travel through matter is much longer than their penetration into a material [4].

2.2 The Photoelectric effect

The photoelectric effect dominates the interaction of low-energy photons (0-0.5MeV) with matter. A photon excites an electron bound to an atom, producing an energetic photoelectron. The minimum amount of energy required to eject an electron from its respective atom is called the work function, Φ, of that material. Thus, for a photon to eject an electron from a material it must have a minimum frequency known as the threshold frequency ν_0. The probability of photoelectric effects occurring is known as the photoelectric cross section, approximately given by

$$\tau = \frac{Z^N}{(hv)^{3.5}} \tag{1}$$

Where τ is the photoelectric cross section, Z is the atomic number of the interacting material, N varies between 4 and 5, h is the plank constant and v is the frequency of the photon. Equation 1 shows that as the photon energy decreases, the probability of photoelectric absorption increases rapidly. The photoelectric interaction is therefore most likely to occur if the energy of the incident photon is just greater than the work function of the material with which it interacts [5].

2.3 Compton Scattering

The Compton scattering effect dominates mid-energy interactions. In Compton Scattering a photon undergoes an inelastic collision with an electron, and is deflected by an angle θ with respect to its original direction as shown in fig 1. The scattering results in a decrease in the photon's frequency as a portion of its energy is transferred to the recoil electron. The momentum of the incident photon is given by

$$p_{\text{photon}} = \frac{E}{c} = \frac{hf}{c} = \frac{h}{\lambda} \tag{2}$$

where p is momentum, E is energy, c is the speed of light, h is planck's constant and λ is wavelength.

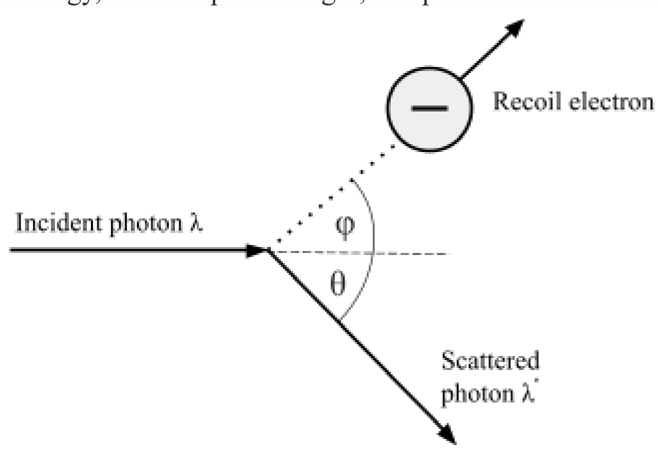

Figure 1: Diagram showing the collision of a photon and an electron at rest in Compton scattering.

Since momentum is conserved in the collision and the photons velocity cannot be reduced, the photon has an increased wavelength after the collision. The shift in wavelength increases with scattering angle and is given by

$$\Delta\lambda = \lambda' - \lambda = \frac{h}{m_e c}(1-\cos(\theta)) \tag{3}$$

Where λ is the wavelength of the initial photon, λ' is the wavelength of the scattered photon, m_e is the mass of the electron and θ is the scattering angle. Since all angles of scattering are possible, the scattering will result in a range of photon energies. The maximum energy change will occur when the scattering angle is 180 degrees. This maximum energy change is given by

$$E_{\text{Compton}} = E_T(\text{max}) = \frac{2E^2}{m_e c^2 + 2E} \tag{4}$$

Where E is the initial energy of the photon. It is impossible for the photon to transfer any more energy than this. Therefore on a graph of a decay spectrum, there is a sharp cutoff known as the Compton edge at E_T(max) [6].

2.4 Pair Production

The third main interaction of photons and matter is pair production. This process dominates for photon energies larger than 10 MeV. A photon passing near the nucleus of an atom will be subjected to strong field effects from the nucleus and can split into an electron-positron pair. Energy and momentum are conserved in this interaction. The threshold energy for pair production to occur is given by the sum of the rest mass energies of an electron and a positron, equal to 1.022 MeV. The remaining energy is converted to the kinetic energy of the electron and positron, given by

$$hv = 1.022MeV + KE_{e^-} + KE_{e^+} \tag{5}$$

where *v* is the frequency of the photon, KE_{e^-} is the energy of the electron and KE_{e^+} is the energy of the positron. The positron quickly recombines with an electron in a process called annihilation, producing two photons with energy of 0.511MeV moving in opposite directions [7].

These are the three major modes of the interaction of photons with matter, photons can also interact with matter through coherent scattering and photodisintegration. However the probability of these effects occurring is much lower than the three major phenomena so they were ignored in this experiment.

3. Experimental details

The detector used in this experiment is the X100-7 silicon photodiode, a semiconductor that converts photons into an electric current. The active area of the diode is a square of area 100 mm^2.[8] The photodiode used is a planar device of the PIN type, which consists of a wide, undoped intrinsic semiconductor sandwiched between a p-type semiconductor and an n-type semiconductor, which are heavily doped [9]. The diode is operated under a reverse bias, creating an extensive (0.9mm thick) region depleted of charge. The advantage of using a PIN diode is that the depletion region exists almost entirely within the intrinsic region, which has a constant width regardless of disturbances applied to the diode. The dark current of a photodiode is the relatively small electric current that flows even when no photons are entering the device, caused by the random generation of electrons and holes in the device. The dark current of the X100-7 is typically 1.5nA [8] and was ignored in the experiment.

When a photon of sufficient energy strikes the diode, an electron hole pair is created via the photoelectric effect. If this occurs in the diodes depletion region, the charge carriers are swept from the junction by the built in electric field of the device. Electrons move towards the cathode and holes move towards the anode, producing a photocurrent. This current can be amplified and used to produce a signal pulse proportional to the energy deposited in the detector. The diode is covered with light blocking black epoxy encapsulant to minimise the photocurrent generated by light photons, increasing sensitivity to radiation. The photon response of X100-7 is shown in figure 2. Absorption probability decreases with increasing energy and is below 2% at 100 KeV, meaning the graphs obtained in the experiment will be heavily biased towards photons with energies of around 1-100 KeV, the energy at which photoelectric absorption and compton scattering dominate photon interactions.

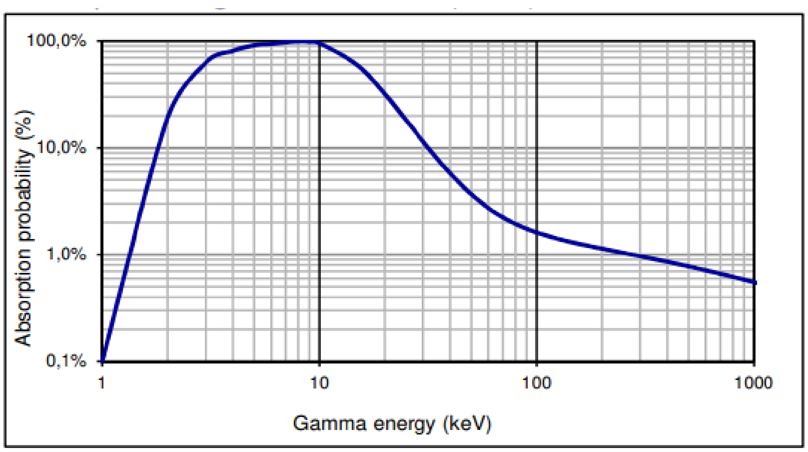

Figure 2: Graph showing absorption probabilities at different photon energies for X100-7 photodiode at 23°C [10].

In the first part of the experiment the detector was exposed to gamma radiation from Am-241, Cs-137 and Na-22 sources. 20mm of perspex was placed between the source and the detector to ensure no beta particles were simultaneously entering the detector. The spectrums of energies for each source were plotted, and noticeable features of the graphs were identified.. An energy calibration was performed using the known energies of features of the decay spectrum for Am-241 and Na-22.

4. Results

4.1 Am-241 decay spectrum

Am-241 decays mainly through alpha decay into Neptunium-237. A gamma ray by product of the decay is produced, which has an 84.6% probability of having energy 59.54 ± 0.01 KeV [11]. This gamma ray results in a clear photopeak at channel number 24 ± 1 as shown in figure 3. This was the first point used in the energy calibration.

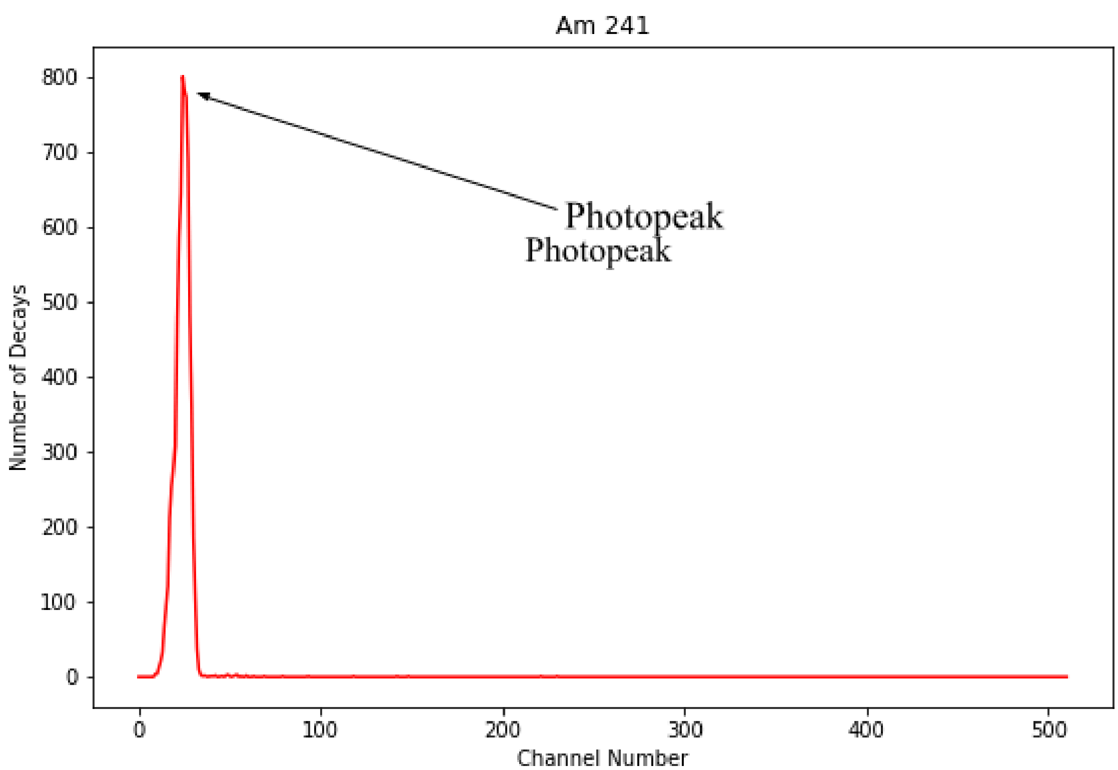

Figure 3: Decay Spectrum of Am-241, taken over a period of 649 seconds.

4.2 Na-22 decay spectrum

The main decay mode of Na-22 is β^+ decay. A positron of energy 546 KeV is emitted with a probability of 90.326% [11]. The emitted positrons annihilate with surrounding electrons, producing two gamma rays of energy 511KeV which undergo compton scattering. The spectrum of Na-22 decay is shown in figure 4.

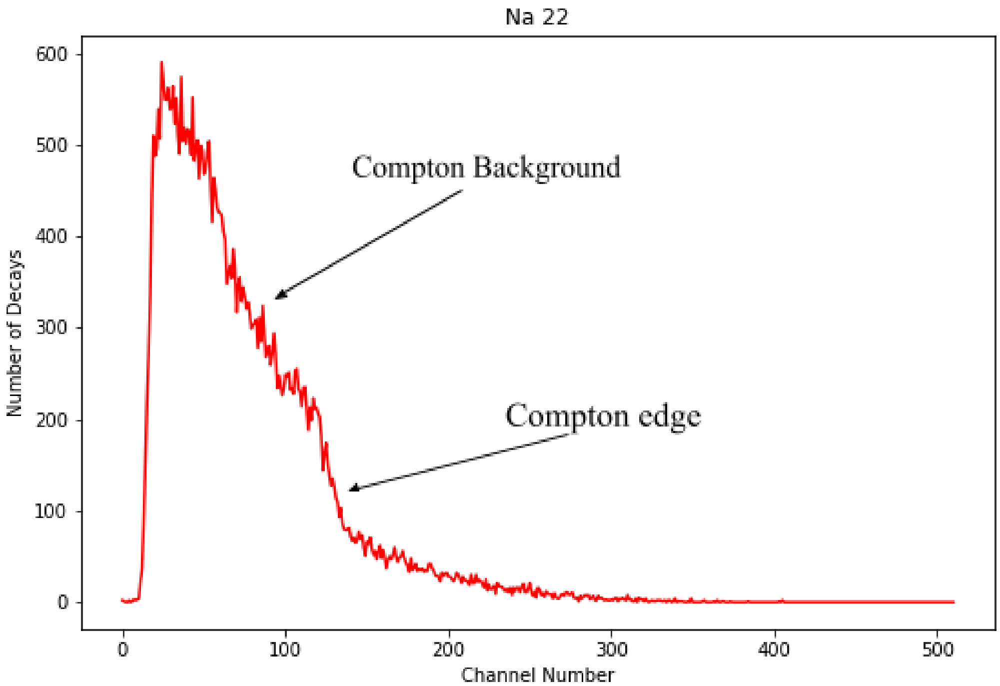

Figure 4: Decay spectrum of Na-22, taken over a period of 4523 seconds.

The compton edge was identified by eye to be channel number 130 ± 10. The gradients of the compton background and compton edge were compared to check there was a noticeable change. Between channel numbers 30 and 120 the average gradient of the graph was -4.26 ± 0.01, between channel numbers 120 and 140 the average gradient was -6.85 ± 0.01, a 61% change. The energy of the Compton edge was calculated using equation 4, the photons were taken to have initial energy 511 ± 1 KeV. $E_T(max)$ was found to be 340.67 ± 0.01 KeV. The formula used for the energy calibration was that of a straight line,

$$E = nG + C \tag{6}$$

where E is energy, n is channel number, and G and C are constants. Using the values for energies and channel numbers from Am-241 and Na-22, G was found to be 2.65 ± 0.08. C was found to be -4.12 ± 0.02 KeV, meaning there is an energy offset. The error on G was calculated in quadrature using the formula,

$$\delta G = G\sqrt{(\frac{\delta E}{E})^2 + (\frac{\delta n}{n})^2} \tag{7}$$

5. Conclusion

In conclusion the photopeak of the Am-241 and the Compton Edge of the Na-22 spectrum were used to perform an energy calibration. The calibration was found to be linear with a low energy offset. The main source of error in this experiment was systematic and came from identifying where the Compton edge of the Na-22 spectrum started. A large uncertainty in the Compton edge channel number was included to try take this into account. A better way of identifying where the Compton edge starts, for example finding the channel number with the greatest change in gradient, would reduce the uncertainty on G. The experiment could also be improved by using a detector with a scintillating crystal, increasing absorption probabilities for high energy gamma rays and allowing further features of the decay spectrum to be identified.

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